Lab Assignment 06

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**Inheritance**
Number of Tasks:	**12 (Classwork: 06, Homework: 06)**

[Submit all the Coding Tasks (Homework: Task 1 to 5) in the Google Form shared on buX before the next lab.]

[You are not allowed to change the driver codes of any of the tasks]

CLASSWORK

Task 1

Complete the class **Circle** so that the desired outputs are generated properly.

Given Code	Expected Output

`public class shapeTester {`
`  public static void main(String[] args) {`
`  Shape s = new Shape();`
`  s.name = "Mobius Strip";`
`  s.color = "Blue";`
`  s.displayInfo();`
`  System.out.println("1======================");`
`  Circle c = new Circle();`
`  System.out.println("2======================");`
`  c.name = "Circle";`
`  c.color = "Red";`
`  c.radius = 5;`
`  c.displayInfo();`
`  System.out.println("3======================");`
`  c.area();`
`  }`
`}`
 	
`public class Shape {`
`  public String name;`
`  public String color;`
 	
`  public void displayInfo() {`
`  System.out.printf("Name: %s\nColor: %s\n", name, color);`
`  }`
`}`
 	
`public class Circle extends Shape {`
`  //Your Code Here`
`}`
	`Name: Mobius Strip`
	`Color: Blue`
	`1======================`
	`2======================`
	`Name: Circle`
	`Color: Red`
	`3======================`
	`Area of Red Circle: 78.54`

Task 2

Given the following classes, write the code for the **BBAStudent** class so that the following output is printed when we run the TestStudent class.

Driver Code and Parent Class	Output

`public class TestStudent{`
`  public static void main(String [] args){`
`  BBAStudent b1 = new BBAStudent();`
`  BBAStudent b2 = new BBAStudent("Humty Dumty");`
`  BBAStudent b3 = new BBAStudent("Little Bo Peep");`
`  b1.details();`
`  System.out.println("1----------");`
`  b2.details();`
`  System.out.println("2----------");`
`  b3.details();`
`  }`
`}`
 	
`public class Student{`
`  private String name = "Just a Student";`
`  private String department = "nothing";`
 	
`  public void setDepartment(String dpt){`
`  this.department = dpt;`
`  }`
`  public void setName(String name){`
`  this.name = name;`
`  }`
`  public void details(){`
`  System.out.println("Name : " + name + " Department: "`
`+ department);`
`  }`
`}`
	`Name: Default Department: BBA`
	`1----------`
	`Name: Humty Dumty Department: BBA`
	`2----------`
	`Name: Little Bo Peep Department: BBA`

Task 3

Given the following classes, write the code for the **Vehicle2010** class to print the following output when we run the Vehicle2010User class.

Driver Code and Parent Class	Output

`public class Vehicle2010User{`
`  public static void main(String[] args){`
`  Vehicle2010 car1 = new Vehicle2010();`
`  System.out.println(car1.getPosition());`
`  car1.moveLowerLeft();`
`  System.out.println(car1.getPosition());`
`  Vehicle2010 car2 = new Vehicle2010();`
`  System.out.println(car2.getPosition());`
`  car2.moveUpperRight();`
`  System.out.println(car2.getPosition());`
`  car2.moveLowerRight();`
`  System.out.println(car2.getPosition());`
`  }`
`}`
 	
`public class Vehicle{`
`  public int x;`
`  public int y;`
`  public void moveUp(){`
`  y = y+1;`
`  }`
`  public void moveDown(){`
`  y = y-1;`
`  }`
`  public void moveLeft(){`
`  x = x-1;`
`  }`
`  public void moveRight(){`
`  x = x+1;`
`  }`
`  public String getPosition(){`
`  return "("+ x + ","+ y + ")";`
`  }`
`}`
	(0,0)
	(-1,-1)
	(0,0)
	(1,1)
	(2,0)

Task 4

Given the following classes, write the code for the **Cricket_Tournament** and the **Tennis_Tournment** classes derived from **Tournament** so that the following output is generated.

Given Code	Expected Output

`public class TournamentTester {`
`  public static void main(String[] args) {`
`  Cricket_Tournament ct1 = new Cricket_Tournament();`
`  System.out.println(ct1.info());`
`  System.out.println("----------");`
 	
`  Cricket_Tournament ct2 = new Cricket_Tournament("IPL", 10, "t20");`
`  System.out.println(ct2.info());`
`  System.out.println("----------");`
 	
`  Tennis_Tournament tt = new Tennis_Tournament("Roland Garros", 128);`
`  System.out.println(tt.info());`
`  }`
`}`
 	
`public class Tournament {`
`  private String name;`
`  public Tournament() {`
`  this.name = "Default";`
`  }`
`  public Tournament(String name) {`
`  this.name = name;`
`  }`
`  public String getName(){`
`  return "Tournament Name: "+name;`
`  }`
`}`
	`Cricket Tournament Name: Default`
	`Number of Teams: 0`
	`Type: No type`
	`----------`
	`Cricket Tournament Name: IPL`
	`Number of Teams: 10`
	`Type: t20`
	`----------`
	`Tennis Tournament Name: Roland Garros`
	`Number of Players: 128`

Task 5

Design the **Dog** and **Cat** class derived from the Animal class with appropriate attributes and properties so that the driver code can generate the output given below.

Driver Code and Parent Class	Output

`public class AnimalTester1 {`
`  public static void main(String[] args) {`
`  Dog dog = new Dog("Buddy", 5, "Brown", "Bulldog");`
`  Cat cat = new Cat("Kitty", 3, "White", "Persian");`
`  System.out.println("1.========");`
`  System.out.println(dog.dogInfo());`
`  System.out.println("2.========");`
`  System.out.println(cat.catInfo());`
`  System.out.println("3.========");`
`  dog.makeSound();`
`  System.out.println("4.========");`
`  cat.makeSound();`
`  }`
`}`
 	
`public class Animal {`
`  public String name;`
`  public int age;`
`  public String color;`
`  public Animal(String name, int age, String color) {`
`  this.name = name;`
`  this.age = age;`
`  this.color = color;`
`  }`
`  public String info() {`
`  return "Name: "+name+"\nAge: "+age+"\nColor: "+color+"\n";`
`  }`
`}`
	`1.========`
	`Name: Buddy`
	`Age: 5`
	`Color: Brown`
	`Breed: Bulldog`
	`2.========`
	`Name: Kitty`
	`Age: 3`
	`Color: White`
	`Breed: Persian`
	`3.========`
	`Brown color Buddy is barking`
	`4.========`
	`White color Kitty is meowing`

Task 6

```
public class A{
  public int temp = 4;
  public int sum = 1;
  public int y = 2;
  public void methodA(int m, int n){
    int x = 0;
    y = y + m + (temp++);
    x = x + 2 +  n;
    sum = sum + x + y;
    System.out.println(x + " " + y+ " " + sum);
  }
}
public class B extends A {
  public int x = 6;
  public void methodB(int m, int n){
   int  y = 0;
   y = ++y + this.y;
   x = this.y + 2 + temp;
   methodA(x, y);
   sum = x + y + super.sum;
   System.out.println(x + " " + y+ " " + sum);
  }
}
```

```
A a1 = new A();
a1.methodA(1, 1);
B b1 = new B();
b1.methodB(1, 2);
```

x	y	sum

HOMEWORK

Task 1

Complete the class **Cow** so that the desired outputs are generated properly.

Given Code

```
public class AnimalTester2 {
  public static void main(String args[]){
    Animal a1 = new Animal();
    System.out.println("1-------------");
    a1.details();
    System.out.println("2-------------");
    Cow c1  = new Cow();
    c1.name = "Pammy";
    System.out.println("3-------------");
    System.out.println("Name: " + c1.getName());
    c1.details();
    System.out.println("4-------------");
    c1.updateSound("Moo");
    System.out.println("5-------------");
    c1.details();
  }
}

public class Animal{
  public int legs = 4;
  public String sound = "Not defined";

  public void details(){
    System.out.println("Legs: "+legs);
    System.out.println("Sound: "+sound);
  }
}

public class Cow extends Animal{
    //Your Code Here
}
```

Expected Output

```
1-------------
Legs: 4
Sound: Not defined
2-------------
The cow says hello!
3-------------
Name: Pammy
Legs: 4
Sound: Not defined
4-------------
5-------------
Legs: 4
Sound: Moo
```

Task 2

Design the **CheckingAccount** class derived from the Account class with appropriate attributes and properties so that the driver code can generate the output given below.

Driver Code and Parent Class	Output

`public class Account{`
`  public double balance = 0.0;`
 	
`  public Account(double balance){`
`  this.balance = balance;`
`  }`
`  public double showBalance(){`
`  return balance;`
`  }`
`}`
 	
//Tester Class
`public class TestAccount{`
`  public static void main(String [] args){`
`  System.out.println("Total Checking Accounts: "+CheckingAccount.count);`
`  CheckingAccount c1 = new CheckingAccount();`
`  System.out.println("Account Balance: " + c1.showBalance());`
`  CheckingAccount c2 = new CheckingAccount(100.0);`
`  System.out.println("Account Balance: " + c2.showBalance());`
`  CheckingAccount c3 = new CheckingAccount(200.0);`
`  System.out.println("Account Balance: " + c3.showBalance());`
`  System.out.println("Total Checking Accounts: "+CheckingAccount.count);`
`  }`
`}`
	`Total Checking Accounts: 0`
	`Account Balance: 0.0`
	`Account Balance: 100.0`
	`Account Balance: 200.0`
	`Total Checking Accounts: 3`

Task 3

Given the following classes, write the code for the **Book** and the **CD** class so that the following output is printed.

Driver Code and Parent Class	Expected Output

`public class Tester6 {`
`  public static void main(String[] args) {`
`  Book book = new Book(1, "The Alchemist", 500, "97806", "HarperCollins");`
`  System.out.println(book.printDetail());`
`  System.out.println("----------");`
 	
`  CD cd = new CD(2, "Shotto", 300, "Warfaze", 50, "Hard Rock");`
`  System.out.println(cd.printDetail());`
`  }`
`}`
 	
`class Product {`
`  private int id;`
`  private String title;`
`  private int price;`
 	
`  public Product(int id, String title, int price) {`
`  this.id = id;`
`  this.title = title;`
`  this.price = price;`
`  }`
 	
`  public String getIdTitlePrice() {`
`  return "ID: " + id + " Title: " + title + " Price: " + price;`
`  }`
`}`
	`ID: 1 Title: The Alchemist Price: 500`
	`ISBN: 97806 Publisher: HarperCollins`
	`----------`
	`ID: 2 Title: Shotto Price: 300`
	`Band: Warfaze Duration: 50 minutes`
	`Genre: Hard Rock`

Task 4

Design the **ComplexNumber** class with the necessary property to produce the output from the given driver code.

Driver Code and Parent Class	Output

`public class ComplexNumberTester {`
`    public static void main(String[] args) {`
`        ComplexNumber cn1 = new ComplexNumber();`
`        System.out.println(cn1.details());`
`        System.out.println("----------");`
`        ComplexNumber cn2 = new ComplexNumber(5.0, 7.0);`
`        System.out.println(cn2.details());`
`    }`
`}`
 	
`public class RealNumber {`
`    public double realValue;`
`    public RealNumber() {`
`        this(0.0);`
`    }`
`    public RealNumber(double realValue) {`
`        this.realValue = realValue;`
`    }`
`    public String getReal(){`
`        return "RealPart: " + realValue;`
`    }`
`}`
	`RealPart: 1.0`
	`ImaginaryPart: 1.0`
	`----------`
	`RealPart: 5.0`
	`ImaginaryPart: 7.0`

Task 5

Given the following classes, write the code for the **CSEStudent** class derived from **Student** so that the following output is generated.

Given Code

```
public class StudentTester{
  public static void main (String args[]){
    CSEStudent.details();
    System.out.println("1--------------");
    CSEStudent s1 =  new CSEStudent("Bob", 23);
    s1.info();
    System.out.println("2--------------");
    CSEStudent s2 =  new CSEStudent("Don", 33);
    s2.info();
    System.out.println("3--------------");
    s1.addLabBasedCourse("CSE220");
    s1.addLabBasedCourse("CSE221");
    System.out.println("4--------------");
    s1.info();
    System.out.println("5--------------");
    CSEStudent.details();
    System.out.println("6--------------");
    s1.addLabBasedCourse("CSE230");
    System.out.println("7--------------");
    s1.info();
    System.out.println("8--------------");
    s2.addLabBasedCourse("CSE110");
    s2.info();
  }
}

class Student{
  public String name;
  public int id;
  public String courses = "";

  public Student(String n, int i){
    name = n;
    id = i;
  }

  public void info(){
    System.out.println("Name: "+name);
    System.out.println("ID: "+id);
    System.out.println("Courses: "+courses);
  }
}
```

Expected Output

```
Total CSE Students: 0
Available Lab Based Courses:
CSE110 CSE111 CSE220 CSE221
1--------------
Name: Bob
ID: 23
Courses:
2--------------
Name: Don
ID: 33
Courses:
3--------------
4--------------
Name: Bob
ID: 23
Courses: CSE220 CSE221
5--------------
Total CSE Students: 2
Available Lab Based Courses:
CSE110 CSE111 CSE220 CSE221
6--------------
It is not a lab based course!
7--------------
Name: Bob
ID: 23
Courses: CSE220 CSE221
8--------------
Name: Don
ID: 33
Courses: CSE110
```

Task 6

```
public class A{
 public int m = 14, n = -1, p = 3;
 public static int temp = 11;
 public void methodA(int m, int n){
   int x = 0;
   p = this.p + m + (this.temp++);
   x = x + 2 +  n;
   this.n = n + B.x + p;
   System.out.println(this.m + " " + p+ " " + this.n);
 }
}
public class B extends A {
 public static int x = 6;
 public void methodB(int x, int y){
  y = ++y + this.p + n;
  super.p = x + 2 + temp;
  methodA(temp, x);
  B.x = this.x + x + super.temp;
  System.out.println(B.x + " " + y+ " " + A.temp);
 }
}
```

```
B b1 = new B();
b1.methodB(3, 5);
A a1 = new A();
a1.methodA(12, -9);
```

x	y	sum

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Given the following classes, write the code for the **Player** and the **Manager** classes derived from SportsPerson class so that the following output is printed. To calculate the match earnings use the following formula:

1. Player: (total_goal * 1000) + (total_match * 10)
2. Manager: match_win * 1000

Given Code	Expected Output

`public class PlayerTester {`
`  public static void main(String[] args) {`
`  Player playerOne = new Player("Al-Nassr", "Ronaldo", "Striker", 25, 32);`
`  playerOne.calculateRatio();`
`  playerOne.printDetails();`
`  System.out.println("----------");`
`  Manager managerOne = new Manager("Real Madrid", "Zidane", "Manager", 25);`
`  managerOne.printDetails();`
`  }`
`}`
 	
`class SportsPerson {`
`  private String team;`
`  private String name;`
`  public String role;`
`  public double earningPerMatch;`
 	
`  public SportsPerson(String teamName, String name, String role){`
`  this.team = teamName;`
`  this.name = name;`
`  this.role = role;`
`  this.earningPerMatch = 0;`
`  }`
 	
`  public String getNameTeam() {`
`  return "Name: " + name + ", Team Name: " + team;`
`  }`
`}`
	Name: Ronaldo, Team Name: Al-Nassr
	Team Role: Striker
	Total Goal: 25, Total Played: 32
	Goal Ratio: 0.78
	Match Earning: 25320K
	---------- 	
	Name: Zidane, Team Name: Real Madrid
	Team Role: Manager
	Total Win: 25
	Match Earning: 25000K